

Fundamentals of PYTHON

Day 1 :

1. The print statement
2. Comments
3. Python Data Structures & Data Types
4. String Operations in Python
5. Python keywords
6. Python Variables
7. Python Type Conversions
8. Simple Input & Output

Print Statement:

The print statement in Python is used to display information on the screen.

Syntax: `print(expression)`

Example:

```
print("Hello, World!")
```

Comments:

Comments in Python are used to explain code and are not executed.

Single-line comments start with `#`, and

multi-line comments are enclosed in triple quotes (`'''` or `"""`).

```
# This is a single-line comment
```

```
'''
```

```
This is a  
multi-line comment  
'''
```

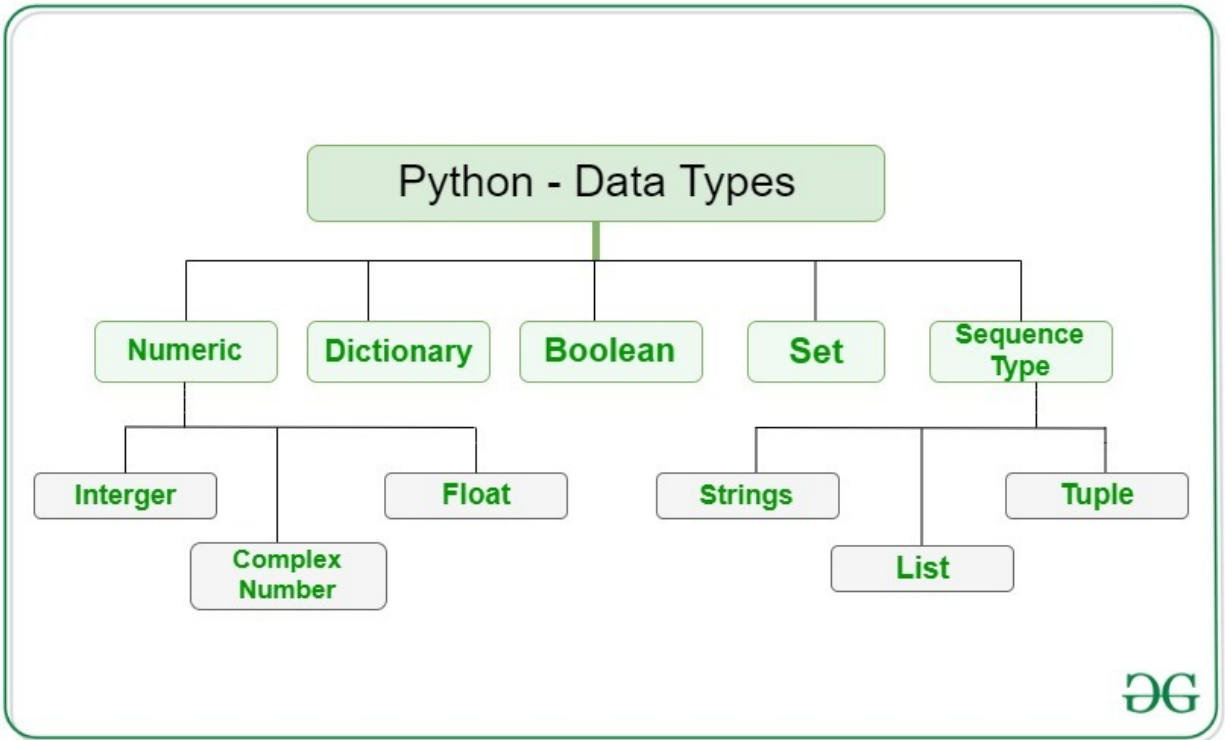
Python Data Structures & Data Types:

Python has various data structures like lists, tuples, sets, and dictionaries. Data types include int, float, str, bool, etc.

Example:

```
# List  
my_list = [1, 2, 3]  
  
# Tuple  
my_tuple = (4, 5, 6)  
  
# Set  
my_set = {7, 8, 9}  
  
# Dictionary  
my_dict = {'a': 10, 'b': 20}  
  
# Integer  
my_int = 42  
  
# String  
my_str = "Hello"  
  
# Boolean  
my_bool = True
```

Data Structures:



Data Types:

1. int:

- Stands for integer.
- Represents whole numbers without any decimal point.

Example:

```
x = 5
```

float:

- Stands for floating-point number.
- Represents real numbers with a decimal point.

Example:

```
y = 3.14
```

str:

- Stands for string.
- Represents a sequence of characters.
- Strings are enclosed in single (`' '`) or double (`" "`) quotes or triple quotes (`''' '''`)

Example:

```
name = "Alice"
```

bool:

- Stands for boolean.
- Represents either `True` or `False` .

Example:

```
is_student = True
```

list:

- As mentioned earlier, a list is a mutable, ordered sequence.

Example:

```
numbers = [1, 2, 3, 4, 5]
```

tuple:

- A tuple is an immutable, ordered sequence.

Example:

```
coordinates = (3, 5)
```

set:

- A set is an unordered collection of unique elements.

Example:

```
unique_numbers = {1, 2, 3, 4, 5}
```

dict:

- A dictionary is an unordered collection of key-value pairs.

Example:

```
person = {"name": "Bob", "age": 30, "is_student": False}
```

String Operations in Python:

Strings can be concatenated using +, repeated using *, and indexed/sliced.

Example:

```
str1 = "Hello"
str2 = "World"

# Concatenation
result = str1 + " " + str2  # "Hello World"

# Repetition
repeated_str = str1 * 3  # "HelloHelloHello"
```

Python Keywords:

Keywords are reserved words that have special meanings in Python.

Examples: if, else, for, while, def, class, return, etc.

36 keywords

```
['False', 'None', '__peg__parser__', 'True', 'and',  
'as', 'assert', 'async', 'await', 'break', 'class',  
'continue', 'def', 'del', 'elif', 'else', 'except',  
'finally', 'for', 'from', 'global', 'if', 'import',  
'in', 'is', 'lambda', 'nonlocal', 'not', 'or',  
'pass', 'raise', 'return', 'try', 'while', 'with',  
'yield']
```

Variables:

- Python Variable is containers which store values.
- Python is not “statically typed”.
- We do not need to declare variables before using them or declare their type.
- A variable is created the moment we first assign a value to it.
- A Python variable is a name given to a memory location.
- It is the basic unit of storage in a program.

Rules for creating variables in Python:

- A variable name must start with a letter or the underscore character.
- A variable name cannot start with a number.
- A variable name can only contain alpha-numeric characters and underscores (A-z, 0-9, and _).
- Variable names are case-sensitive (name, Name and NAME are three different variables).
- The reserved words(keywords) cannot be used naming the variable.

Variable Cases:

There are 3 types of variable cases in python:

- 1.Camel Case
- 2.Snake Case
- 3.Pascal Case

Simple Input & Output:

`input()` is used to take user input, and `print()` is used to display output.

Example:

```
user_name = input("Enter your name: ")  
print("Hello, " + user_name + "!")
```